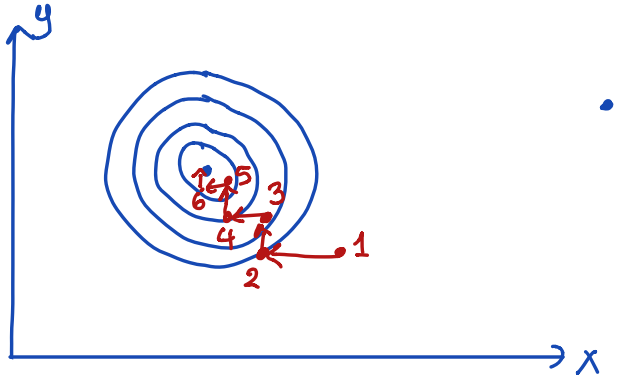
Multidimensional Optimization

(Ch. 14 in the book "Numerical Methods for Engineers")

Univariate Search Method

- Change one variable at a time to improve the approximation while the other variables are held constant.
- Problem reduces to a sequence of 1D searches

Graphically:

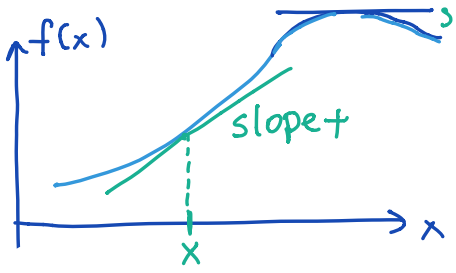


Start at pt. 1

- Move along x-axis with y constant
- Next move along y-axis with x constant
- Repeat the above 2 steps.

Gradient Methods

In 1D: First derivative of a 1D function provides a slope or tangent to the function.

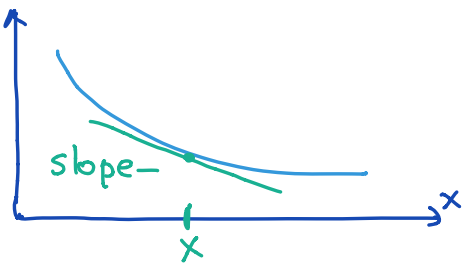


⇒ If slope > 0, increasing x will increase f(x)

⇒ If slope = 0, we are at optimum

$f'(x) = 0$

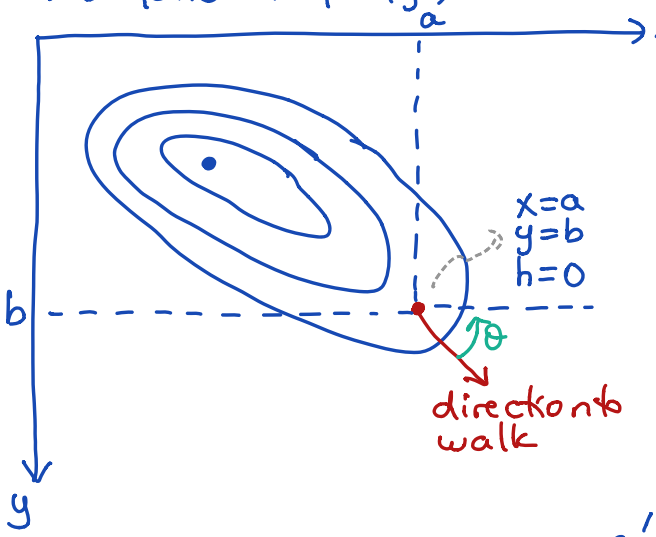
- if $f''(x) > 0 \Rightarrow$ a minimum (concave up)
- if $f''(x) < 0 \Rightarrow$ a maximum (concave down)



⇒ If slope < 0, decreasing x will increase f(x)

The Gradient

2D function $f(x,y)$



- You are on a mountain and want to reach the top
- You can move in arbitray direction
- What's the slope in that direction.
- The elevation along the direction (red arrow)

Directional derivative

$$g'(0) = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$

evaluated at $x=a$, $y=b$

- The elevation along direction (red arrow) can be thought of a new function

$g(h)$. Slope at $x=a, y=b$

$g'(0)$

Goal: To gain the most elevation with the next step, what's the direction of steepest ascent?

Gradient: $\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j}$ or $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$

"del f"

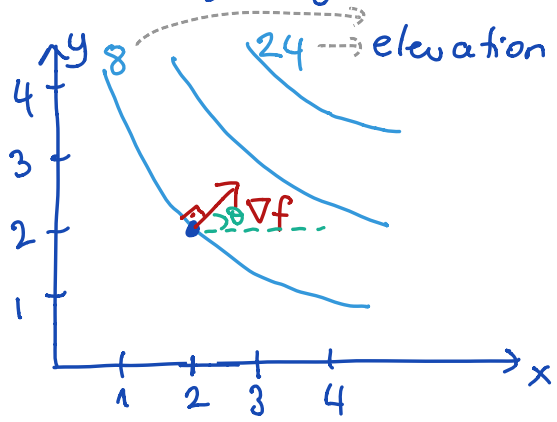
Generalize to n-dimensions:

$$\nabla f(\underline{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\underline{x}) \\ \frac{\partial f}{\partial x_2}(\underline{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\underline{x}) \end{bmatrix}$$

(*) Gradient tells us what direction to go locally for gaining max. elevation and how much we will gain by taking it.

EX Employ the gradient to evaluate the steepest ascent direction for the function:

$f(x,y) = xy^2$ at the point $(2,2)$



Gradient (∇f) is orthogonal to the elevation contour.

Sol Evaluate elevation:

$f(2,2) = 2(2)^2 = 8$

Find the partial derivatives:

$$\frac{\partial f}{\partial x} = y^2\Big|_{y=2} = 4 \qquad \frac{\partial f}{\partial y} = 2xy\Big|_{\substack{x=2\\y=2}} = 8$$

$$\text{Gradient} = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

$\nabla f = 4\underline{i} + 8\underline{j}$, $\theta = \tan^{-1}\left(\frac{8}{4}\right) \cong 63^\circ$

slope in that direction: $\|\nabla f\| = \sqrt{4^2+8^2} = 8.944$

we will gain 8.944 units of elevation for a unit distance travelled along the steepest path (red arrow)

Note: As we move forward, the direction and magnitude of the steepest path will change. You can modify your climbing at each step.